A Sai Karthik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.55
Narayana Junior College	Class XII (TELANGANA BOARD OF INTERMEDIATE EDUCATION)	2024	97.90%
Narayana Concept School	Class X (TELANGANA BOARD OF SECONDARY EDUCATION)	2022	9.30%

PROJECTS

Duolingo -- Monetization vs. Engagement

- Analyzed Duolingo's 2025 monetization strategy and 2026 strategic reversal using **public company disclosures and user-sentiment data** to identify a **segmentation mismatch** affecting highly engaged free users.
- Designed **"Streak Mastery,"** a **segment-targeted product intervention** for high-accuracy, long-streak users, with an **A/B testing framework** covering revenue-retention trade-offs.
- Developed an **evidence-based business trade-off framework** to evaluate when sacrificing **short-term monetization for long-term retention** is justified.

Fair Batch -- Quick Commerce Dispatch

- Investigated the potential link between **order-clubbing and rider late-delivery penalties**, using **public regulatory, labor, and industry evidence** to frame a rider-fairness hypothesis.
- Designed **"Fair Batch,"** a **transparent, rule-based dispatch and SLA system** that attributes delays by cause before applying rider penalties, alongside a **rider-facing prototype**.
- Built an **A/B experimentation and metrics framework** with **Fair Delivery Rate** as the North Star and **delivery-time/cost guardrails** to evaluate the business trade-off.

C++ HTTP/1.1 Server | *C++17, POSIX Sockets, Linux, epoll*

- Developed an HTTP/1.1 server using low-level TCP socket programming with non-blocking, event-driven I/O.
- Implemented HTTP request parsing, connection management, request handling, and HTTP response generation.
- Designed the server architecture around epoll for handling multiple concurrent client connections efficiently.

Neural Network From Scratch | *Python, NumPy, Machine Learning*

- Implemented a feed-forward neural network from scratch using NumPy without deep-learning frameworks.
- Implemented forward propagation, ReLU activation, softmax, cross-entropy loss, and manual backpropagation.
- Applied mini-batch gradient descent and gradient checking to train and validate the network on MNIST.

SCHOLASTIC ACHIEVEMENTS

Cleared National Talent Search Examination **(NTSE)**, securing a scholarship offered by **Govt. of India.**

POSITIONS OF RESPONSIBILITY

Events Coordinator | Milan, IIT Hyderabad

Coordinated end-to-end execution of 6 sports across 23 hostels, maintaining clash-free schedules and recording match results for timely web updates.

Infrastructure Coordinator | Elan & nVision, IIT Hyderabad

Conducted and coordinated ~10 technical workshops across AI/ML, cybersecurity and data science, with individual sessions reaching **250+ students**.

Transport Coordinator | Inter-IIT Sports Meet, IIT Hyderabad

- Coordinated transportation for contingents from **23 IITs** and staff, managing **train bookings, bus allocation, and arrival schedules** to ensure smooth movement from arrival points to IITH.
- Planned and coordinated **transport to rented external sports grounds**, working with travel agencies to allocate buses by capacity and maintain centralized schedules for timely movement.

Public Relations & Networking Coordinator | Media Council, IIT Hyderabad

Managed and maintained **ECHO IITH**, handling regular content publishing and promotion of IITH events, hackathons, and student initiatives in coordination with the Media Council team.

SKILLS

Product: Design Thinking, User-Centric Design, Wireframing & Prototyping, Market Research, Competitive Strategy, Problem Framing, Hypothesis-Driven Thinking, Prioritization, Trade-off Analysis

AI: Generative AI, LLM Fundamentals, Prompt Engineering

Data & Analytics: SQL, Product Analytics, A/B Testing, Excel, Power BI

Programming: Python, C++

RELEVANT COURSES

Relevant Courses: Fundamentals of Machine Learning, Database Management Systems(DBMS), Data Structures & Algorithms(DSA), Probability, Statistics, Linear Algebra

EXTRACURRICULAR

Team Manager | IITH's Volleyball League, IIT Hyderabad

Managed both **men's and women's volleyball teams**, building squads through player bidding and leading **both teams to the semifinals**.

Online Tutor | TeacherOn

Taught **Mathematics and Physics for IIT-JEE, BITSAT and SAT** to 4 students over 2 years through personalized online instruction.